AERIS-10 — FREQUENCY_SYNTHESIZER — Schematic sheet 1/1

Drawing ID: ELEC-SCH-FS-S1 Revision: A Date: 2026-10-09 Units: mm Scale: 1:1 at the stated page size
Source: 4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.sch (EAGLE 9.6.2 XML; file revision: none recorded in the source) Status: SOURCE-DERIVED
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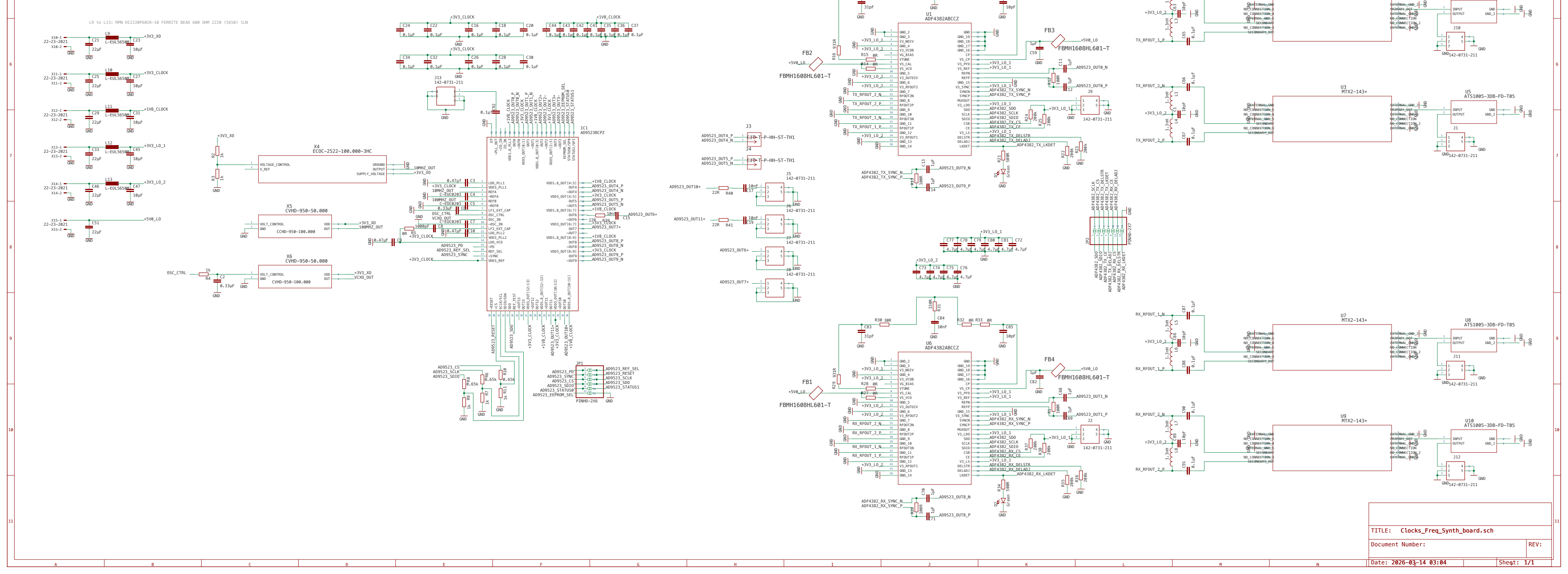
AD9523 OUT	Frequency	Format	Destination	Divider
OUT0	300 MHz	LVDS	ADF4382 #TX REF (REFP/REFN)	/12
OUT1	300 MHz	LVDS	ADF4382 #RX REF (REFP/REFN)	/12
OUT4	400 MHz	LVDS	ADC (AD9484) CLK	/9
OUT5	400 MHz	LVDS	FPGA (phase-aligned copy of ADC clock)	/9
OUT10	120 MHz	LVCMOS	DAC	/30
OUT11	120 MHz	LVCMOS	FPGA (paired with DAC)	/30
OUT6	100 MHz	LVCMOS	FPGA system clock	/36
OUT7	X (e.g., 50 MHz)	LVCMOS	Test clock	/72 for 50 MHz (or pick any)
OUT9	60 MHz	LVDS	ADF4382 #TX SYNC (SYNCP/SYNCN)	60
OUT8	60 MHz	LVDS	ADF4382 #RX SYNC (SYNCP/SYNCN)	Same as OUT9

Jitter Cleaner, Clock Generator & Frequency Synthesizer


IC1 AD9523BCPZ
X4 ECOC-2522-100.000-3HC
X5 CVHD-950-50.000
X6 CVHD-950-50.000
J13 142-0731-211
JP1 PINHD-2X6
JP2 PINHD-2X7
U1 ADF4382ABCCZ
U6 ADF4382ABCCZ
FB1 FBMH1608HL601-T
FB2 FBMH1608HL601-T
FB3 FBMH1608HL601-T
FB4 FBMH1608HL601-T
U2 MTX2-143+
U3 MTX2-143+
U7 MTX2-143+
U9 MTX2-143+
U4 ATS1005-3DB-FD-T05
U5 ATS1005-3DB-FD-T05
U8 ATS1005-3DB-FD-T05
U10 ATS1005-3DB-FD-T05
J2 142-0731-211
J9 142-0731-211
J10 142-0731-211
J11 142-0731-211
J12 142-0731-211
+3V3_XO
+3V3_CLOCK
+1V8_CLOCK
+3V3_LO_1
+3V3_LO_2
+5V0_LO
TX_RFOUT_1_N
TX_RFOUT_1_P
TX_RFOUT_2_N
TX_RFOUT_2_P
RX_RFOUT_1_N
RX_RFOUT_1_P
RX_RFOUT_2_N
RX_RFOUT_2_P
GND

TITLE: Clocks_Freq_Synth_board.sch
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Date: 2026-03-14 03:04
Sheet: 1/1